Scale and proportional relationships

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Use scale factors and identify relationships with a constant rate per one.

R1. A rectangle is 8 cm by 2 cm. Find its area.

R2. List multiples of 4 and 6. Find the least positive common multiple.

R3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

R4. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

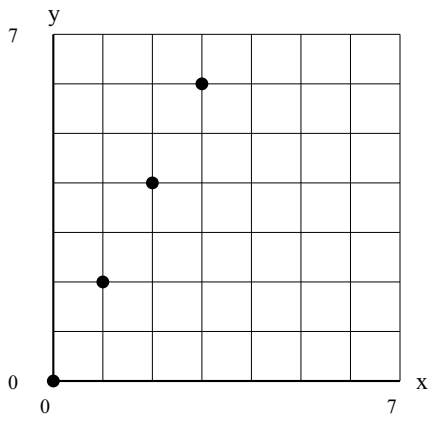
P1. A 4 cm by 6 cm rectangle is copied with scale factor 2. Find the new side lengths and compare areas.

P2. A triangle has side lengths 6, 8, and 10 cm. Copy it with scale factor $\frac{1}{2}$. Find its side lengths. What happens to its angles?

P3. A map uses 1 cm for 25 km. Two towns are 3.6 cm apart on the map. Find their actual distance.

P4. A drawing has area 5 square cm. A scaled copy has all lengths three times as large. Find its area.

Lesson model



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Practice and compare methods

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P5. A table pairs x = 1, 2, 4 with y = 2.5, 5, 10. Find k and write y = kx.

P6. Compare two straight-line graphs: A passes through (0, 0), (1, 2), (2, 4). B passes through (0, 3), (1, 5), (2, 7). Which is proportional?

P7. A ride costs $4 to start and $3 per mile. Write cost c for m miles. Is it proportional?

P8. A faucet fills $\frac{3}{4}$ liter in $\frac{1}{2}$ minute at a constant rate. Write liters L after t minutes.

F. A proportional relationship has (x, y) = (4, 14). Complete: k = 14 ÷ __ = __, so y = __x.

D. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

X1. A rectangle 3 cm by 2 cm is scaled by 2. Find new dimensions and area.

X2. A drawing has area 6 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

Worked example - compare with the slide model

A copy of a 3 cm by 5 cm rectangle uses scale factor 2.

The new lengths are 6 cm and 10 cm.

The original area is 15 square cm; the copy area is 60 square cm.

The area factor is 60 ÷ 15 = 4 = 2^2.

Result: 6 by 10 cm; area 60 square cm

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Challenge, explain and check

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C1. A map scale is 2 cm for 15 km. A route has map legs of 3.6 cm and 2.4 cm. Find the real length of each leg and the full route. A second map doubles every map length. What scale should it show for the same route?

C2. A6 cm by 10 cm photo is enlarged by scale factor 1.5. Find the new dimensions, perimeter, and area. Compare the perimeter factor with the area factor. Explain why multiplying the original area by 1.5 is insufficient.

Finish the lesson

S. A proportional relationship has __ for every nonzero input and passes through __.

E. A table pairs x = 1, 3, 5 with y = 5, 11, 17. Is y proportional to x? Support your answer with ratios or the value at zero.